DAFFA AHMAD PANGREKSA

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Driyorejo, Gresik, Jawa Timur

Currently pursuing a Bachelor's degree in Data Science at Surabaya State University. Possesses practical experience in machine learning, natural language processing, and data analysis gained through academic projects, competitions, and active research in a big data laboratory. Proficient in working with Python and various data science frameworks, from data preprocessing through to model deployment.

Education

Universitas Negeri Surabaya (UNESA) - **Surabaya, Indonesia** Jun 2024 - Jul 2028 (Expected)

S1 Data Science, 3.92/4.00

- Relevant Courses: Machine Learning/ AI, Data Mining, Data Wrangling, Exploratory Data Analysis, Programming (Python), Data Structures & Algorithms, Databases, Statistics.

Professional Experience

Data Science and AI Researcher Mar 2026 - Present

Laboratorium Big Data Analysis, FMIPA, UNESA

- Conducting active research in the field of Graph Neural Networks (GNN) for large- scale relational data modeling using Neo4j and NetworkX.
- Developing an experiment pipeline based on Jupyter Notebooks as part of ongoing scientific research outputs.

Teaching Assistant in Data Structure and Algorithm Apr 2026 - Present

Data Science Study Program, FMIPA UNESA

- Mentored 70+ students on the concepts of search algorithms, sorting, recursion, and time and space complexity analysis.
- Led lab sessions and evaluated programming assignments to ensure students' understanding of data structure implementation.

Teaching Assistant in Basic Programming Sep 2025 - Dec 2025

Data Science Study Program, FMIPA UNESA

- Mentored over 70 students from two classes in mastering Python programming logic.
- Led weekly lab sessions and provided structured feedback to improve the technical efficiency of students' code.

Event Division Staff - ACTION Competition 2025 (Organization) Jun 2025 - Dec 2025

HMP Sains Data (HIMASADA) UNESA

- Designed and managed a national- level infographic competition (150+ participants) in a structured and seamless manner.
- Served as the lead moderator, facilitating communication among the three judges, participants, and the organizing committee.
- Accurately managed and compiled the final competition data to ensure transparency in the judging process.

RANK 6 - Data Mining Competition (DISCO) May 2025 - Jun 2025

Universitas Trunojoyo Madura

- Building a profit prediction model for MSMEs using XGBRegressor with Optuna optimization yielded an R^2 of 0.91.
- Conducting percentile-based business segmentation into three profitability tiers from 13,564 MSME data points across various industries.

Logistics and Supplies Division Staff - RELOUD 2024 (Organization) Sep 2024 - Nov 2024

HMP Sains Data (HIMASADA) UNESA

- Managed the logistics for a large-scale event (300+ participants) with 100% equipment readiness and no issues.
- Coordinated with 5+ departments and external vendors, achieving cost efficiencies of approximately 15% within budget.
- Oversaw the distribution, maintenance, and return of equipment using a well-organized inventory system.

Project Experience

Comparative Sentiment Analysis of the MBG Program Using LDA and Fine-Tuned IndoBERT Sep 2025 - Dec 2025

Data Science Study Program, FMIPA, UNESA

- Processed 4,968 tweets and 3,418 YouTube comments via dual-platform scraping and a preprocessing pipeline based on regular expressions, stopword removal, and tokenization.
- Applying LDA topic modeling with coherence score optimization yielded 10 optimal topics on X (score 0.498) and 3 topics on YouTube (score 0.407).
- Classifying sentiment using fine-tuned IndoBERT (average confidence 0.89) revealed that negative sentiment was more dominant on YouTube (49.1%) than on X (46.9%).

Predicting Diabetes Risk in the 2017–2020 NHANES Dataset Sep 2025 - Dec 2025

Data Science Study Program, FMIPA, UNESA

- Integrating 10 XPT files from the CDC API, which include demographic data, BMI, and clinical history, and engineering composite features such as the LAP Index, VAI Score, and Metabolic Risk Score.
- Comparing XGBoost, Random Forest, and MLP; Random Forest achieved the best results after leakage correction with an F1 score of 0.526 and an AUC of 0.895.

Benzene Estimator: Multi-Sensor Air Quality Analysis Dec 2025

Data Science Study Program, FMIPA, UNESA

- Building a Streamlit-based web application for real-time estimation of benzene (C6H6) concentration using sensor array data from the UCI ML Repository.
- Developing an OLS model using statsmodels and scikit-learn, complete with interactive Plotly gauge charts and evaluation metrics including R^2, MAE, and RMSE.
- Implementing an automated preprocessing pipeline based on IQR and interactive EDA with correlation heatmaps and multivariate 3D scatter plots.

Music Genre Classification and Recommendation Systems Based on Digital Signal Processing and Machine Learning Oct 2025 - Nov 2025

Data Science Study Program, FMIPA, UNESA

- Designing a digital signal processing pipeline using a 5th-order Butterworth filter and extracting 58 acoustic features (ZCR, MFCC, Chroma STFT) from the 999-track GTZAN audio dataset.
- Training 4 supervised learning models; the MLP achieved the highest accuracy of 76.50% (F1 = 0.767), with MFCC confirmed as the most discriminative feature (separation ratio 1.55x).
- Building a recommendation engine based on cosine similarity in a 58-dimensional feature space for identifying similar acoustic tracks across genres.

East Java Regional Food Affordability Index Nov 2025 - Dec 2025

Data Science Study Program, FMIPA, UNESA

- Integrating three heterogeneous data sources—UMK documents, Bank Indonesia price data, and Tokopedia scraping—for 11 commodities across 9 cities in East Java.
- Developing the Market Affordability Index (MAI) and E-Commerce Access Index (EAI) normalized using

MinMaxScaler, revealing a 2x purchasing power gap between Surabaya (MAI >120) and Sumenep (MAI ~60).
- Publishing the final dataset publicly on Kaggle as the ‘East Java Affordability Index’.

Certifications and Skills

- **Dicoding Certification**: Data Science; Fundamentals of Artificial Intelligence; Introduction to Financial Literacy
- **Digital Talent Scholarship Certification, Ministry of Communication and Information Technology (Kemenkominfo)**: Data Analyst
- **Programming Languages**: Python (Pandas, NumPy, Scikit-learn, TensorFlow, Statsmodels, Librosa, Pytorch)
- **Machine Learning & AI**: Supervised and Unsupervised Learning, NLP, Graph Neural Network, Deep Learning, Feature Engineering, Feature Selection, Feature Extraction
- **Data & Visualization**: SQL, Streamlit, Plotly, Matplotlib, Seaborn
- **Tools & Platform**: Jupyter Notebook, Git, Neo4j, NetworkX, Figma, Tableau, PowerBI, Microsoft Office
- **Languanges**: Bahasa: Indonesia (Native), Inggris (Intermediate)
- **Soft Skills**: Communication, Teamwork, Leadership, Time Management, Attention to Detail, Problem Solving